ThermoGuardian AI

Smart Continuous Fever Monitoring System
Predicting Fever Before It Becomes Critical

The Problem

- • 250,000+ annual U.S. sepsis deaths linked to delayed detection
  - • Temperature checked every 4–8 hours in most hospitals
  - • 40% of nocturnal fevers go undetected until morning
  - • No continuous solution for general ward patients

The Clinical Gap

- • Fever = early warning sign of sepsis & infections
  - • Monitoring is intermittent and reactive
  - • Nurses spend ~30% of shift on manual vitals
  - • Gap between thermometers and ICU monitors

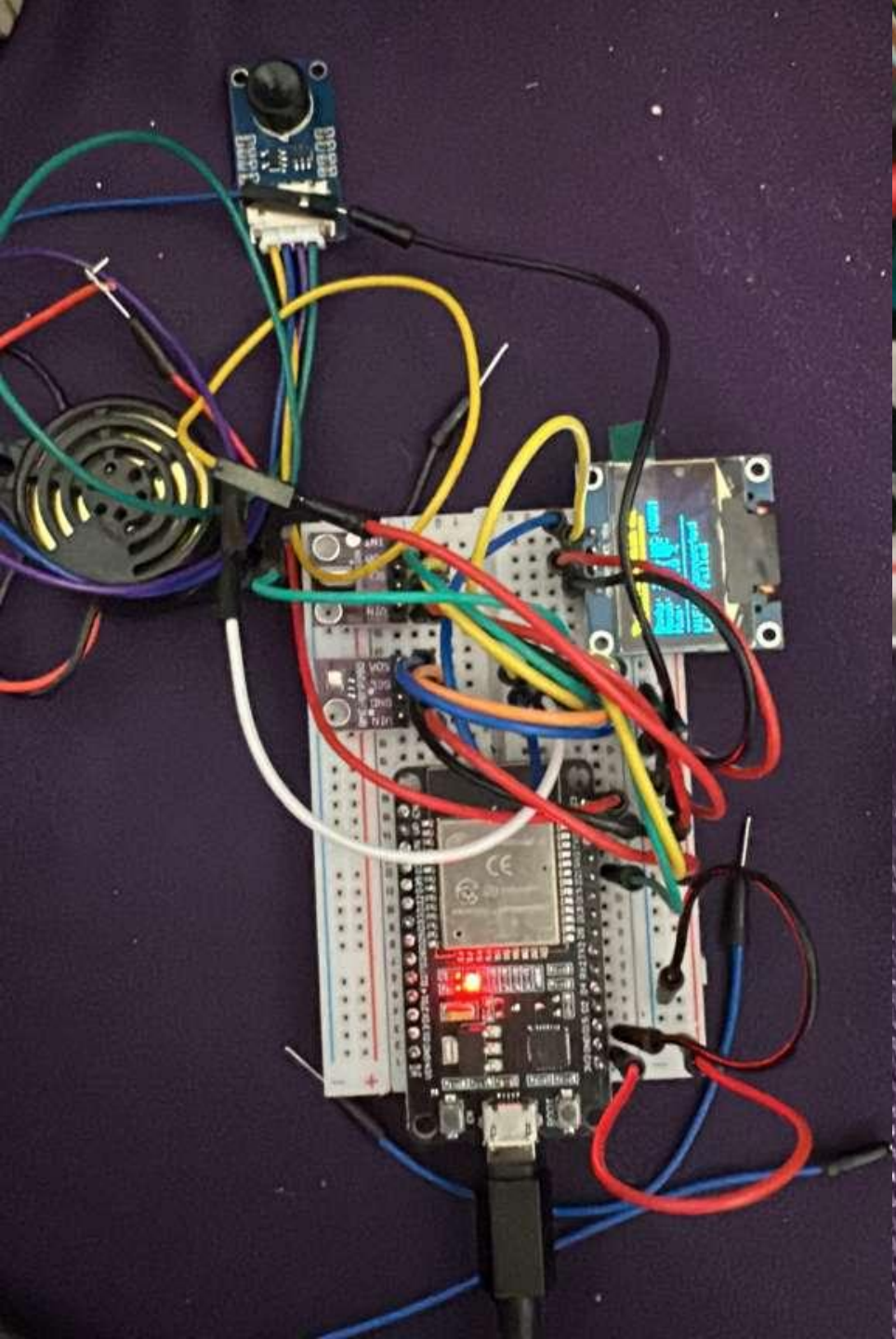
Our Solution

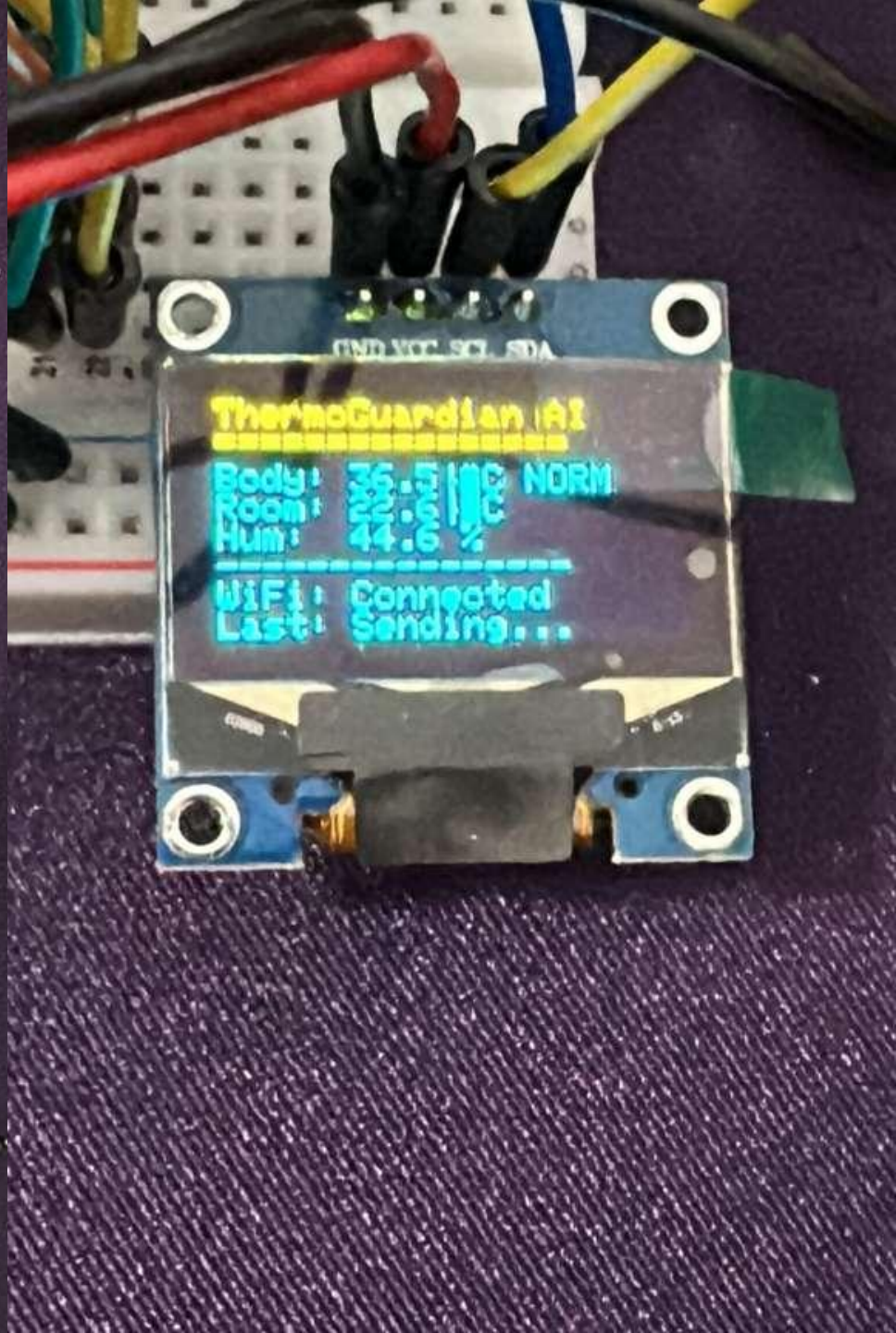
- • Non-contact continuous fever monitoring
  - • Bed-mounted smart device
  - • 24+ hour battery operation
  - • AI-driven predictive analytics
  - • Predicts fever 60–90 minutes early

Core Innovation: Multi-Sensor Fusion

- • Thermal Imaging Array (MLX90640)
  - • Heart Rate & SpO2 Sensor (MAX30102)
  - • Environmental Sensor (BME280)
  - • Generates calibrated “Fever Index”
  - • 95% accuracy vs 72% single-sensor systems


Raspberry Pi 5
Trxcom®
TRJG0926HENL4R
China M 2425




GND VCC SCL SDA
ThermoGuardian AI
Body: 36.5°C NORM
Room: 22.6°C
Hum: 44.6 %
WiFi: Connected
Last: Sending...

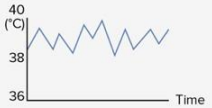
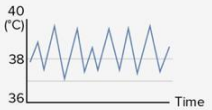
Predictive AI Early Warning

- • Detects rate-of-change patterns
  - • +0.5°C/hour temperature rise
  - • +15% heart rate elevation
  - • Yellow Alert: 60–90 minutes before 38.5°C
  - • Enables proactive intervention

Edge AI Architecture

- • ESP32 microcontroller platform
  - • TensorFlow Lite model on-device
  - • <500ms anomaly detection
  - • Fully operational during WiFi outages
  - • Reliable, fast, independent

Types of fever patterns

Pattern type	Description	Associated diseases
Continuous fever	 Temperature continuously remains above normal, with daily fluctuations < 1°C.	• Typhus • Viral pneumonia
Remittent fever	 Temperature continuously remains above normal, with daily fluctuations ≥ 1°C.	• Typhus • Sepsis • Tuberculosis • Rheumatic fever
Intermittent fever	 Temperature remains above normal only for a certain period, later returning back to normal.	• Pleuritis • Sepsis
Biphasic fever	 Fever that breaks and returns twice a day	• Yellow fever • Dengue • Malaria • Typhoid • Leptospirosis
Undulant fever	 Temperature continuously remains above normal, with daily fluctuations ≥ 1°C.	• Brucellosis
Recurrent fever	 **Relapsing fever:** Days of fever followed by an afebrile period of several days and then a relapse into additional days of fever, usually after 14–21 days. **Pel-Ebstein fever:** Fever lasting 1–2 weeks followed by 1–2 weeks of an afebrile period **Periodic fever:** Fever that recurs over months or years; associated with the absence of infection of malignancy	• Pleuritis • Sepsis

Esp reading the data from pi

Scanning / collecting data

SQLite is for database

Creation of table

API is just like keys provide connections b/w devices

API is used for communication b/w server

@app. Route data. Is used to access data from SQLite / c/

Df= select

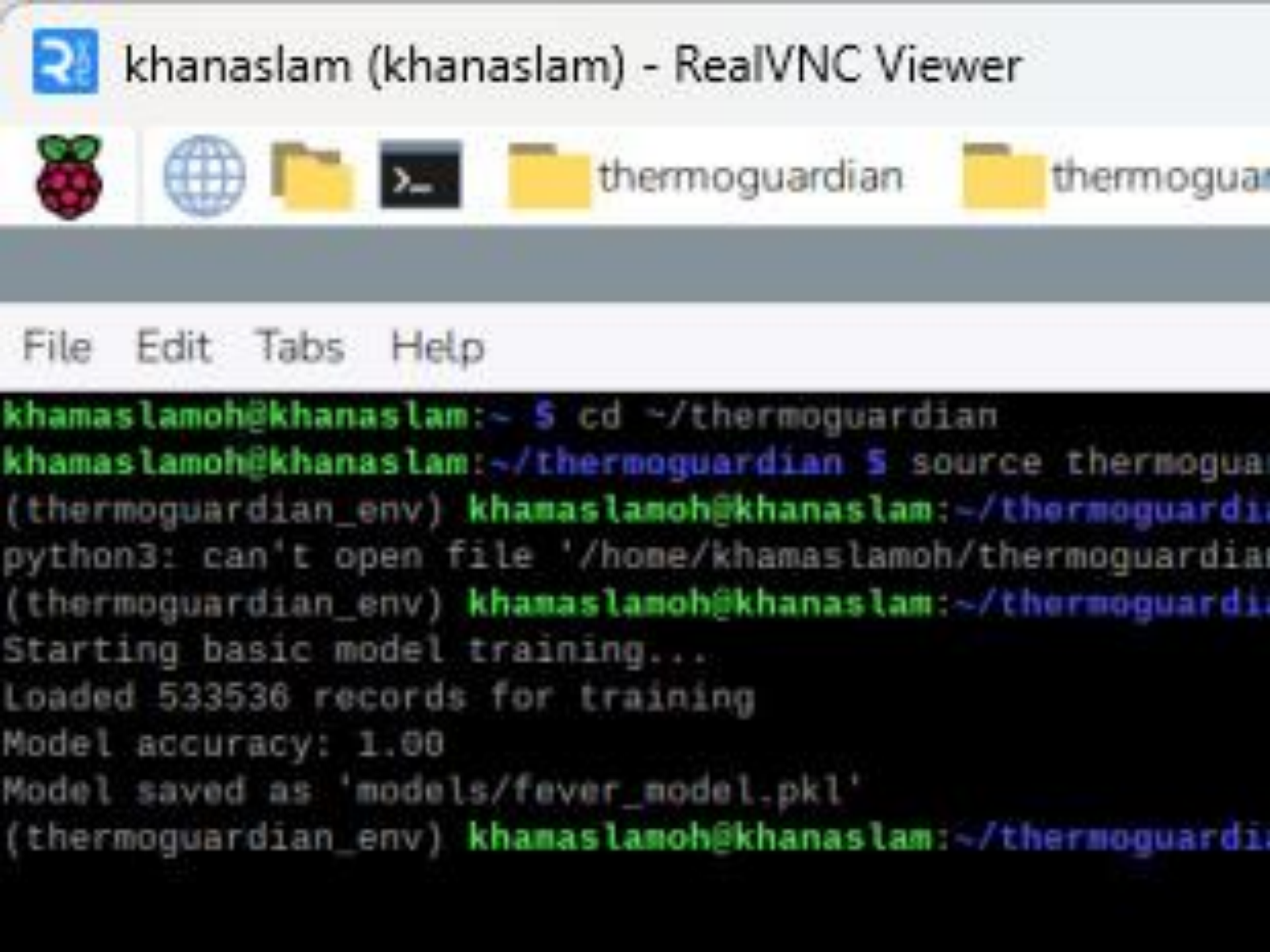
Pd=panda

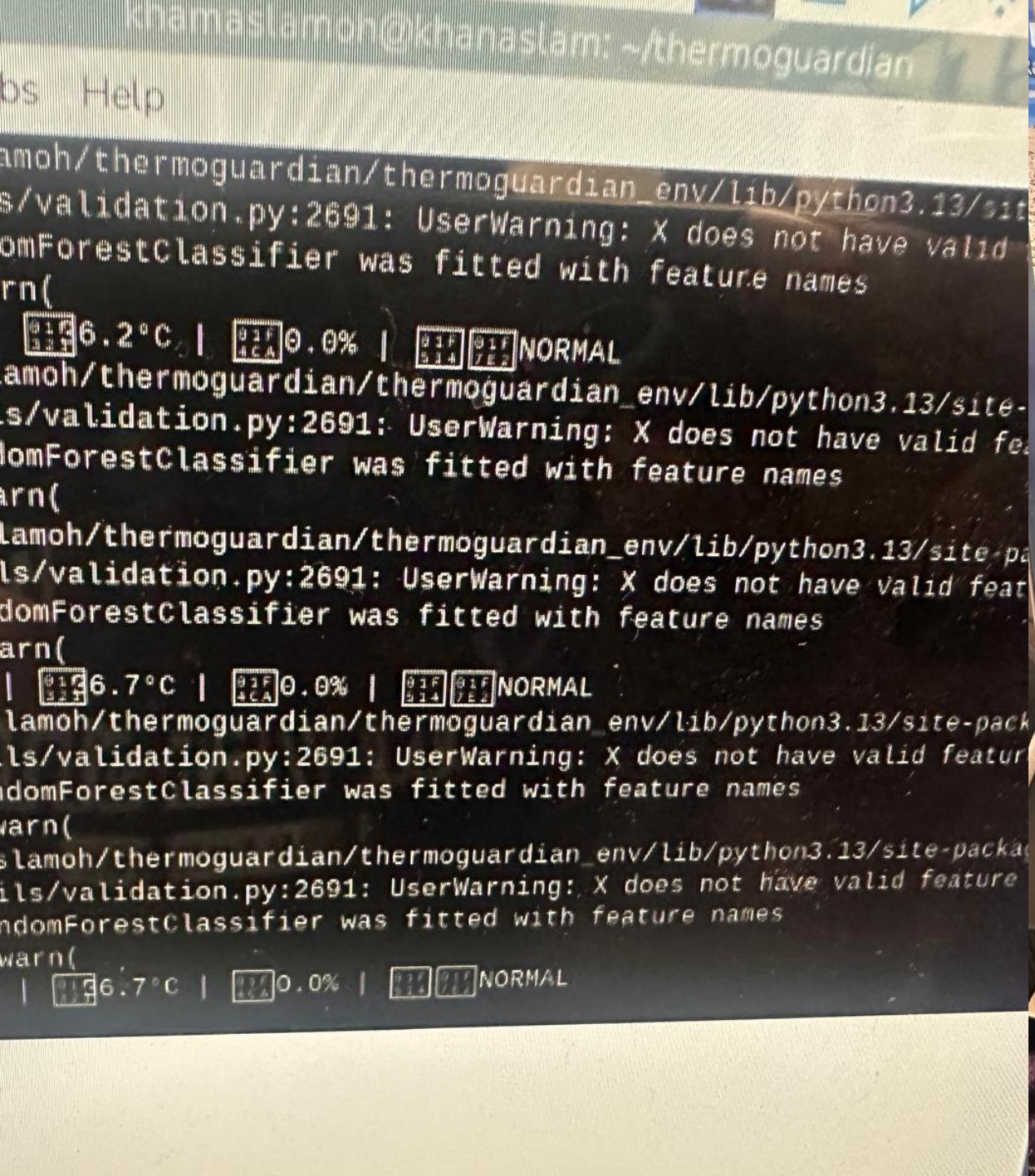
Model training (random forest classifier)

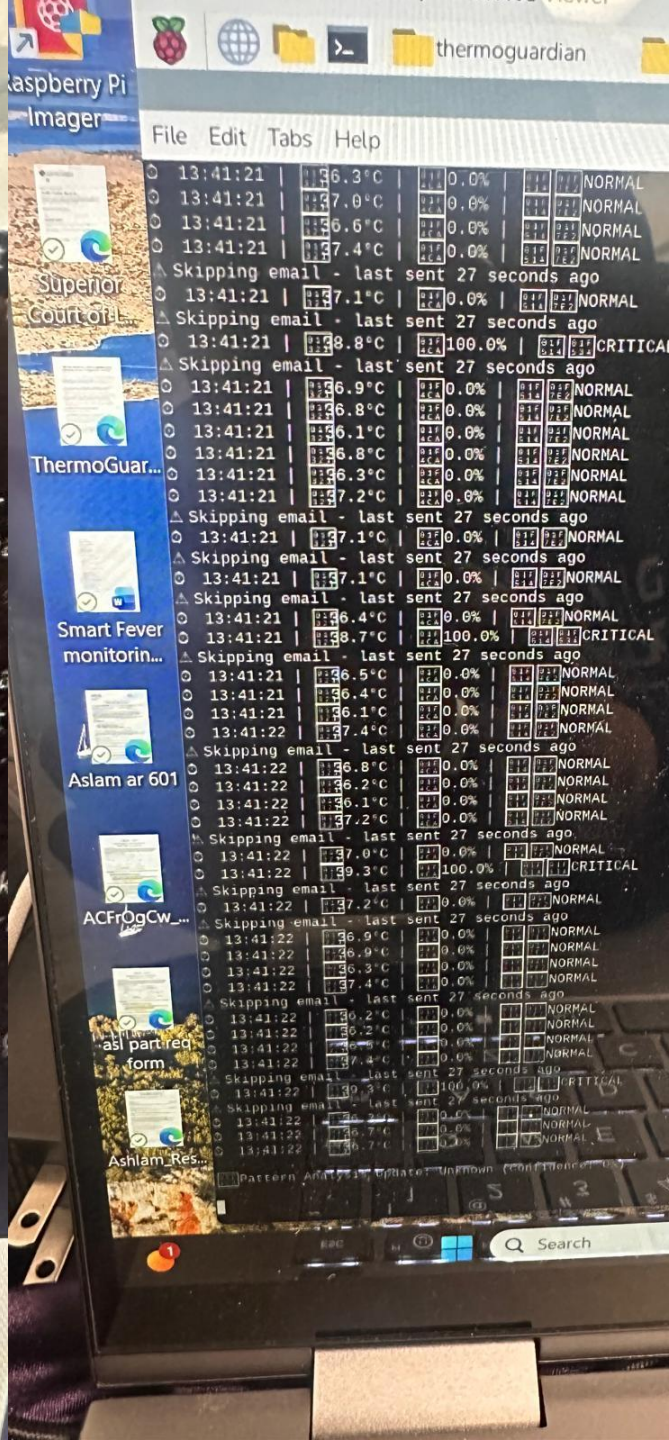
Tree factor looks like tree

Unsupervised method

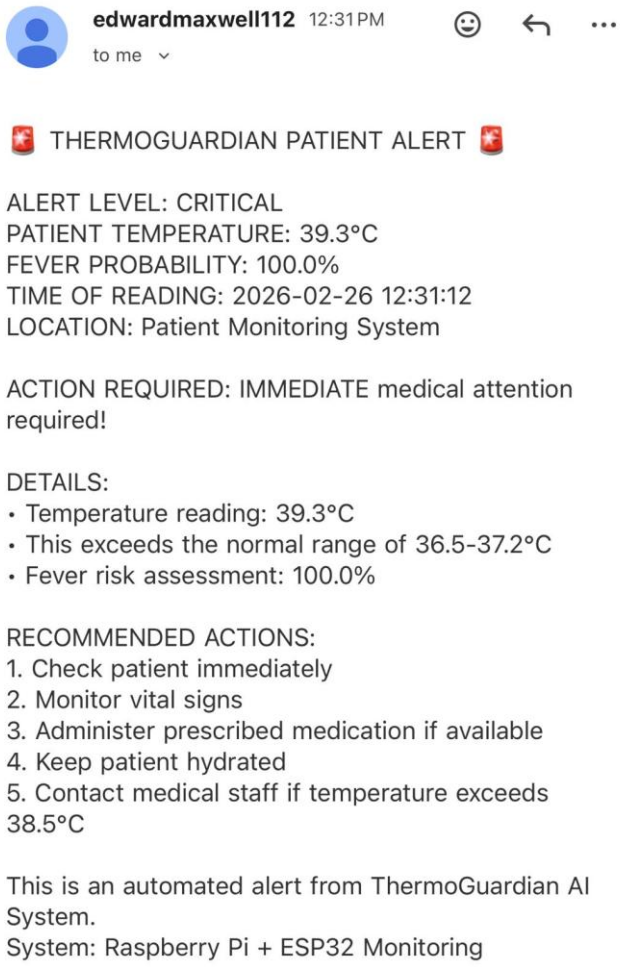
Temperature is calculated in c•


khanaslam (khanaslam) - RealVNC Viewer
thermoguardian
File Edit Tabs Help
khamaslamoh@khanaslam:~ $ cd ~/thermoguardian
khamaslamoh@khanaslam:~/thermoguardian $ source thermogua
(thermoguardian_env) khamaslamoh@khanaslam:~/thermoguardi
python3: can't open file '/home/khamaslamoh/thermoguardia
(thermoguardian_env) khamaslamoh@khanaslam:~/thermoguardi
Starting basic model training...
Loaded 533536 records for training
Model accuracy: 1.00
Model saved as 'models/fever_model.pkl'
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khanaslamon@khanaslam: ~/thermoguardian
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thermoguardian
File Edit Tabs Help
Raspberry Pi Imager
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ThermoGuar...
Smart Fever monitorin...
Aslam ar 601
ACFrOgCw_...
asl part req form
Ashlam_Res...
Skipping email - last sent 27 seconds ago
NORMAL
CRITICAL
Search


edwardmaxwell112 12:31 PM
Patient Temperature Alert: 39.3°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 3 12:30 PM
Patient Temperature Alert: 37.4°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 4 12:26 PM
Patient Temperature Alert: 37.3°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 12:25 PM
Patient Temperature Alert: 38.6°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 12:21 PM
Patient Temperature Alert: 39.1°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 3 12:20 PM
Patient Temperature Alert: 37.2°C
THERMOGUARDIAN PATIENT ALERT A...
edwardmaxwell112 2 12:16 PM
Patient Temperature Alert: 37.1°C


edwardmaxwell112 12:31 PM
to me
THERMOGUARDIAN PATIENT ALERT
ALERT LEVEL: CRITICAL
PATIENT TEMPERATURE: 39.3°C
FEVER PROBABILITY: 100.0%
TIME OF READING: 2026-02-26 12:31:12
LOCATION: Patient Monitoring System
ACTION REQUIRED: IMMEDIATE medical attention required!
DETAILS:
• Temperature reading: 39.3°C
• This exceeds the normal range of 36.5-37.2°C
• Fever risk assessment: 100.0%
RECOMMENDED ACTIONS:
1. Check patient immediately
2. Monitor vital signs
3. Administer prescribed medication if available
4. Keep patient hydrated
5. Contact medical staff if temperature exceeds 38.5°C
This is an automated alert from ThermoGuardian AI System.
System: Raspberry Pi + ESP32 Monitoring

Designed for Clinical Workflow

- • Bed-rail compatible mounting
  - • Automatic positioning
  - • Tiered alerts (Yellow & Red)
  - • Reduces alarm fatigue
  - • Enhances nurse efficiency

Market Opportunity

- • 6,000+ U.S. hospitals
  - • 900,000 hospital beds
  - • 20% target = 180,000 units
  - • $1,200–$1,500 per device
  - • $99 per bed/month software subscription

Roadmap & Clinical Validation

- • Prototype completion (0–3 months)
  - • Volunteer validation & patent filing (4–6 months)
  - • 50-patient hospital pilot (7–12 months)
  - • Target >90% clinical accuracy

Our Vision

- • Detect deterioration before patients crash
  - • Reduce sepsis risk
  - • Improve hospital outcomes
  - • Transform monitoring from reactive to predictive